Dhananjaya Kumar Mittapalli

San Francisco, CA | +1 (857) 832-8906 | dhananjayakumarmittapalli@gmail.com | Linkedin | Github | Portfolio

SUMMARY

Data Engineer with 4+ years building production data pipelines and platforms across finance, retail, and healthcare, specializing in **Python**, **SQL**, **Apache Airflow**, **Apache Spark**, and **Apache Kafka** on **AWS**, **GCP**, and **Azure**. Skilled in designing **Snowflake** and **dbt** data warehouses, streaming and batch **ETL/ELT** pipelines, and **data quality** and **governance** frameworks that scale to millions of records. Proven record of automating manual workflows, cutting pipeline latency and cost, and partnering cross functionally to deliver measurable business impact.

PROFESSIONAL EXPERIENCE

Community Dreams Foundation — San Francisco, CA
Data Engineer — Feb 2026 – Present

- Built and validated **classification models** in **Python** and **SQL** on structured client and deal data for a production **ML** investor ranking system, achieving over **80% accuracy** and reducing deal preparation time through automated, data driven targeting.
- Automated **pricing optimization** workflows end to end in **Python** and **SQL**, building regression and forecasting pipelines with approximately **80% model fit** and reducing manual analysis effort by **70%**.
- Built predictive and causal inference pipelines for price elasticity and demand forecasting, translating raw transactional data into automated, production ready models that contributed to a **15% increase** in sales.
- Designed and deployed **time series forecasting** pipelines to estimate field service parts demand, integrating structured and unstructured data sources and reducing stockouts by **20%**.
- Built and deployed executive **Tableau** dashboards delivering daily, weekly, and monthly operational metrics, working cross functionally with service, operations, and leadership teams on data pipeline requirements.

Natixis Investment Managers — Boston, MA
Data Engineer — Jan 2025 – Aug 2025

- Architected a real time **data warehouse** ingestion layer using **Python** and **Apache Kafka**, streaming 50,000+ trade and AUM events per second with automated schema validation and **data quality SLAs** across 12 production pipelines, reducing data to decision latency by **60%**.
- Designed **dbt** dimensional and **star schema** models on **Amazon Redshift**, standardizing financial KPIs into a governed knowledge layer queried via **Presto/Trino**, eliminating 15+ weekly manual reconciliation hours and cutting reporting cycle time by **40%**.
- Automated **ETL** workflows and orchestration using **Apache Airflow** (Cloud Composer) for revenue pipeline refresh on **GCP**, reducing daily reporting cycles by **40%** while enforcing data quality checks across downstream systems.
- Built an AI augmented **data quality monitoring** system with anomaly severity classification and **Airflow** orchestrated routing, reducing false positive alert noise by **70%** and achieving 100% SLA adherence across critical datasets.
- Defined **event logging** schemas and data governance standards for 8 C suite reporting datasets in **Snowflake**, enabling structured behavioral analytics and full **data lineage** documentation for audit compliance.

SaksOFF5th — Bangalore, India
Business Insight Engineer — May 2022 – Aug 2023

- Engineered a **Snowflake** data warehouse with **dbt** star schema models capturing end to end supply chain KPIs across 500K+ daily orders, and built **Tableau** dashboards enabling leadership to greenlight a 2 day delivery rollout backed by live analytics.
- Productionized an ML driven carrier scoring model in **PySpark** replacing a $200K per year third party tool, cutting fulfillment costs by **10%** and carrier data procurement overhead by **95%**, with model freshness **SLAs** enforced via **Airflow**.
- Designed an AI powered **anomaly detection** pipeline using statistical modeling to flag inventory discrepancies and fraudulent refunds in real time, implementing data quality validation with **99.7% accuracy SLA** across 50+ ingestion feeds and protecting $1.2M in shrink exposure.
- Automated 20+ **ETL** batch pipelines via **Apache Airflow**, reducing report generation from 2 days to 2 hours (**75% faster**), and partnered with data scientists and PMs to define **event logging** schemas enabling **A/B testing** infrastructure for 200K+ weekly active users.

TheMathCompany — Bangalore, India
Data Engineer — Sep 2020 – May 2022

- Led **ETL** data migration from legacy **Aster** and **SAS** databases to **Snowflake** with **AWS S3** staging, designing and maintaining pipelines via **Apache Airflow** on **Docker** and **Kubernetes** hosted on **AWS EC2**, achieving a **140% increase** in pipeline execution speed with less than **2% downtime**.
- Led a cross functional team of 3 to 4 data engineers through strict deadlines, enforcing code review standards and version control discipline, expanding the data engineering practice from 1 to 4 engineers and securing 3 new client engagements.
- Engineered large scale **ETL** pipelines using **PySpark** and **Apache Spark**, optimizing data structures, partitioning, storage strategies, and **Spark Streaming** jobs for distributed computing workloads across multiple client environments.
- Implemented automated **data quality** validation and schema enforcement frameworks across 30+ ingestion pipelines, defining **SLA** contracts achieving **99.7% accuracy** and reducing downstream model retraining failures by **40%**.

EDUCATION

Northeastern University, Boston, MA — Sept 2023 – Dec 2025
Master of Science in Data Science — **CGPA: 3.88 / 4**
Related Courses: Large Language Models, MLOps, Machine Learning, Data Management and Processing, Algorithms, Deep Learning, NLP

M S Ramaiah Institute of Technology, Bengaluru, India — Aug 2016 – Aug 2020
Bachelor of Technology in Electronics and Communication Engineering — **CGPA: 8.83 / 10**
Related Courses: Python Application Programming, Distributed Computing Systems, Data Structures, Digital Signal Processing

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Scala, Java, Bash, Shell Scripting
Data Engineering: Apache Kafka, Apache Airflow, dbt, Apache Spark, Spark Streaming, ETL/ELT design, data warehouse architecture, star schema, dimensional modeling, event logging, data lineage, SLA management, schema evolution
Query Engines: Presto/Trino, Hive, Hadoop, MapReduce, HDFS, Distributed Computing
Databases: Snowflake, Amazon Redshift, BigQuery, PostgreSQL, MySQL, MongoDB, DynamoDB, NoSQL
Cloud: AWS (S3, EMR, EC2, Lambda, Glue, Kinesis, CloudWatch, SageMaker), GCP (BigQuery, Cloud Composer, Cloud Run, Cloud Functions, Dataflow, Vertex AI), Azure (Synapse Analytics, Azure Data Lake, ADF, Databricks, Blob Storage, Azure VM)
Big Data: PySpark, Databricks, Delta Lake, Apache Spark, Kafka, Hadoop, HDFS
AI / LLM: LangChain, LangGraph, LlamaIndex, FAISS, ChromaDB, Pinecone, RAG pipelines, LLM fine tuning, prompt engineering, agent orchestration, Claude API, OpenAI API, Ollama
ML Frameworks: PyTorch, Hugging Face, Scikit learn, XGBoost, TensorFlow, OpenCV, BERT, Sentence BERT
MLOps / DataOps: MLflow, DVC, TFDV, Docker, Kubernetes, GitHub Actions, CI/CD, Travis CI, Terraform
Visualization: Tableau, Power BI, MicroStrategy, Looker, Streamlit, Plotly, Metabase, Adobe Analytics
Practices: Data governance, privacy compliant modeling, responsible AI, bias mitigation, A/B testing infrastructure, data quality SLAs, audit compliance
Other: DataIKU, Fivetran, Rundeck, REST APIs, FastAPI, Modal GPU

PROJECTS

C.O.S.M.I.C | *Python, FastAPI, Modal GPU, WebSocket, Claude Vision* June 2026

- Architected a parallel data ingestion pipeline routing structured panel events to 3 concurrent AI models via a **Modal** serverless GPU orchestration layer, achieving under **500ms end to end latency**.
- Designed a **WebSocket** streaming layer and **FastAPI** orchestration server coordinating parallel model inference; won 1st place Suno Challenge and 3rd place Google DeepMind Challenge among 100+ participants.

Multiclass Disease Classification Pipeline | *Docker, Airflow, DVC, GCP, TFDV, GitHub Actions* Sep 2024 – Dec 2024

- Built a production grade **MLOps** pipeline classifying 14 thoracic diseases from 112,120 chest X-ray images, orchestrating the full model lifecycle with **Docker**, **Apache Airflow** DAGs, **DVC** data versioning, and **GitHub Actions** CI/CD.
- Implemented **TFDV** schema validation as a mandatory CI/CD gate, achieving **90%+** classification accuracy while cutting manual model iteration time by **50%** and reducing Docker memory footprint by **40%**.

Smart City Big Data Analytics Platform | *Apache Spark, Python, Pandas, GBTRegressor* Feb 2024 – Apr 2024

- Engineered distributed **Spark** pipelines processing millions of records across 3 Vancouver urban datasets; trained and compared **Gradient Boosted Trees**, **Random Forest**, and **Linear Regression** models for housing price and CO2 forecasting.

AWARDS & RECOGNITION

- **IEEE Best Paper Award** – Deep Reinforcement Learning for Multi Agent Systems, IEEE MysuruCon 2021, published in IEEE Xplore
- **1st Place, Suno Challenge** – Music HackSpace Hackathon, Berklee College of Music, June 2026 (100+ participants)
- **3rd Place, Google DeepMind Challenge** – Music HackSpace Hackathon, Berklee College of Music, June 2026
- **Best Award, Mini Project Exhibition 2019** – Reinforcement Learning for Legged Locomotion in Robotics, M S Ramaiah Institute of Technology